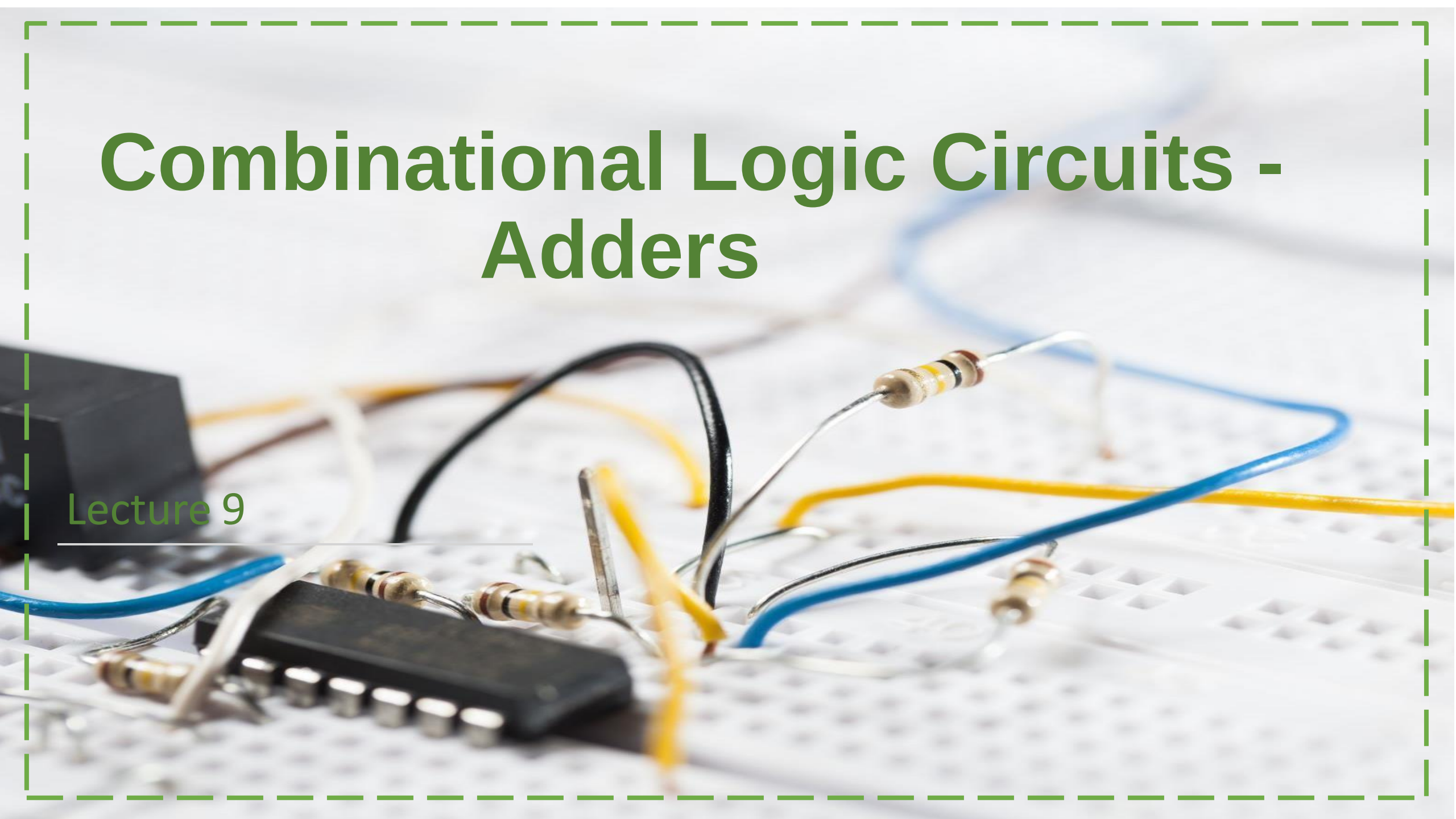
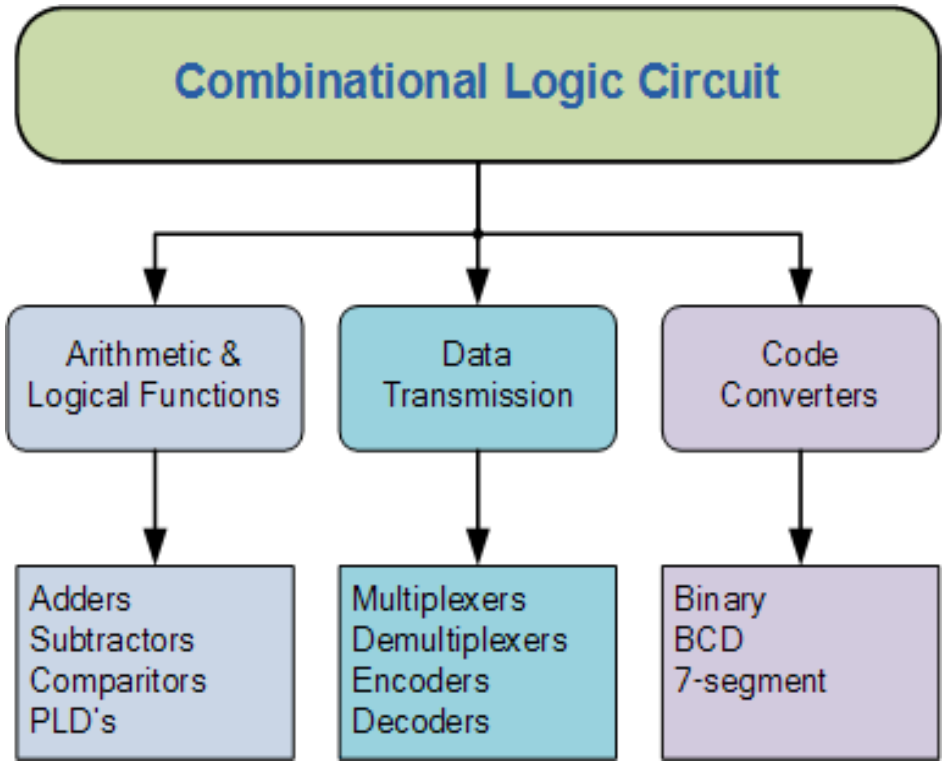
Combinational Logic Circuits - Adders



Lecture 9

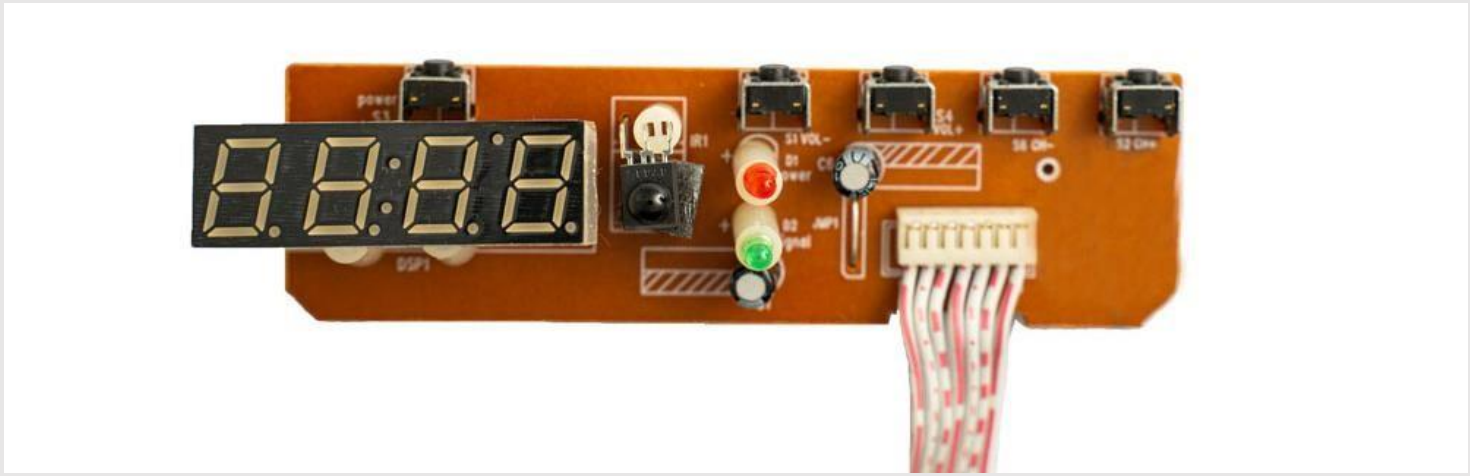
Recall…

- Logic Circuits
  - Combinational
  - Sequential
- Commonly used combinational circuits are fabricated as ready-made ICs.
- No need to construct these circuits from basic logic gates all the time.
- Can use these ICs as building blocks to design complex logic solutions.

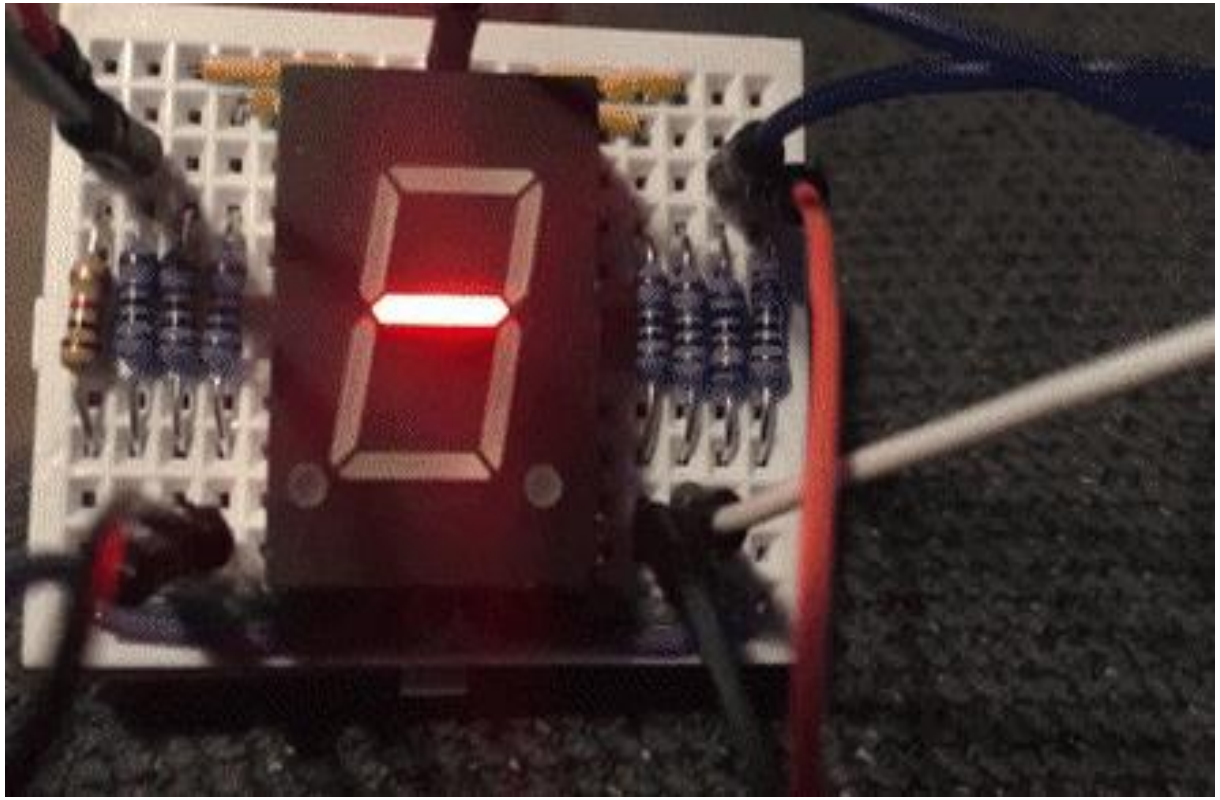


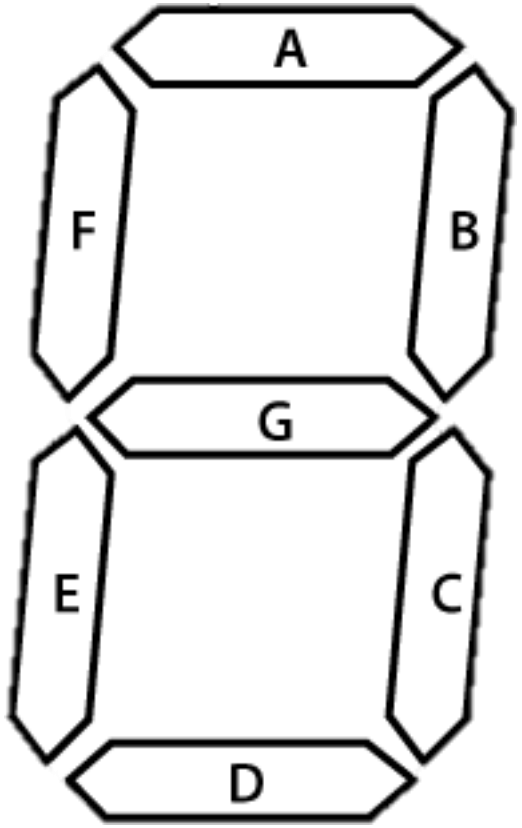
7447-BCD input to SSD

- IC number 7447 are circuits for BCD to seven segment decoder and display from numbers 0 to 9.
- It decodes BCD in seven segmentations.
- Normally, IC7447 accepts input that is binary code values.
- The BCD is processed into seven segments shown as numbers 0 to 9 on a monitor.



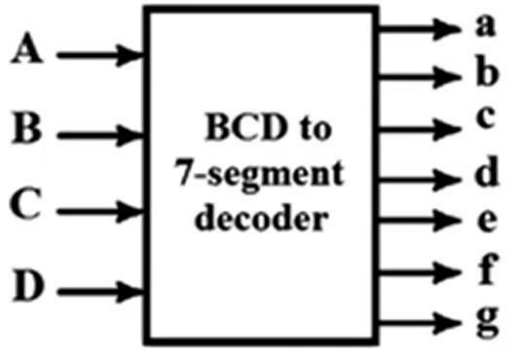
Seven Segment Decoder

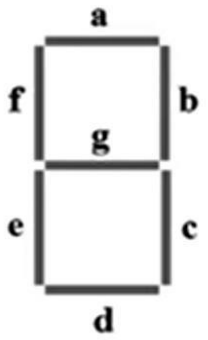




7-Segment Numeric LED Displays

BCD to 7-segment decoder





A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1

https://everycircuit.com/circuit/6711990626287616/binary-to-7-segment-display-decoder

Adders

- An Adder is extensively used for the addition of numbers.
- 3 kinds of Adders
    1. Half Adder
    2. Full Adder
    3. Ripple carry Adder



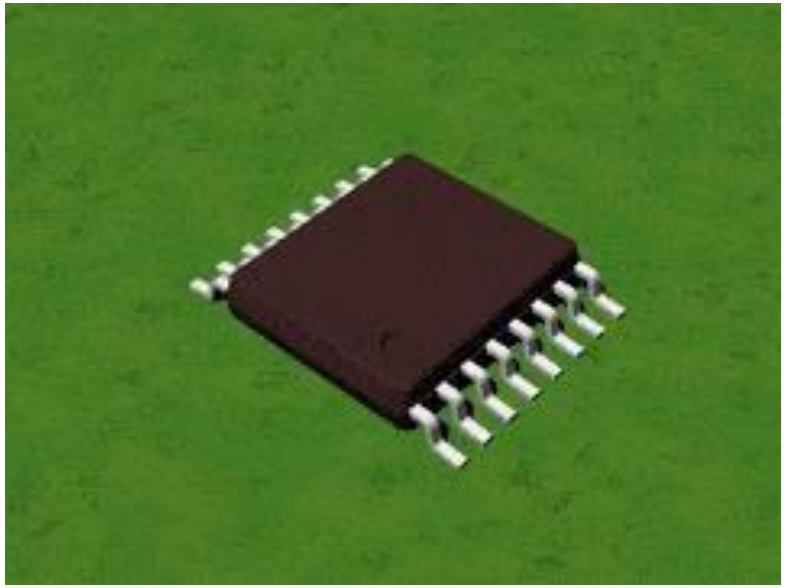
Half adder

- Half Adder is a combinational logic circuit.
- It is used for the purpose of adding two single bit numbers.
- It contains 2 inputs and 2 outputs (sum and carry).



Adder Designing

- Step 1 - Identify the input and output variables-
- Step 2 - Draw the truth table
- Step 3 - Draw K-maps using the above truth table and determine the simplified Boolean expressions
- Step 4 - Draw the logic diagram.



Half Adder Designing



Step 1

- Identify the input and output variables-
  - Input variables = A, B (either 0 or 1)
  - Output variables = S, C where S = Sum and C = Carry

Step 2

Draw the truth table

Inputs		Outputs	
A	B	C (Carry)	S (Sum)
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

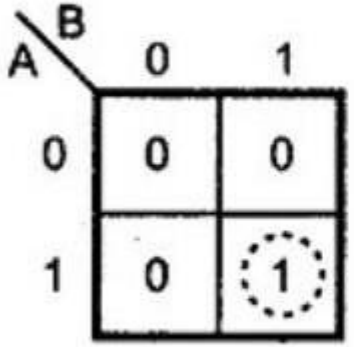
Truth Table

Half Adder Designing

Step 3

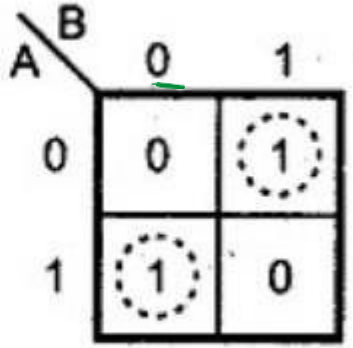
- Draw K-maps using the above truth table and determine the simplified Boolean expressions

For carry



Carry = AB

For Sum



$S = A\overline{B} + \overline{A}B$

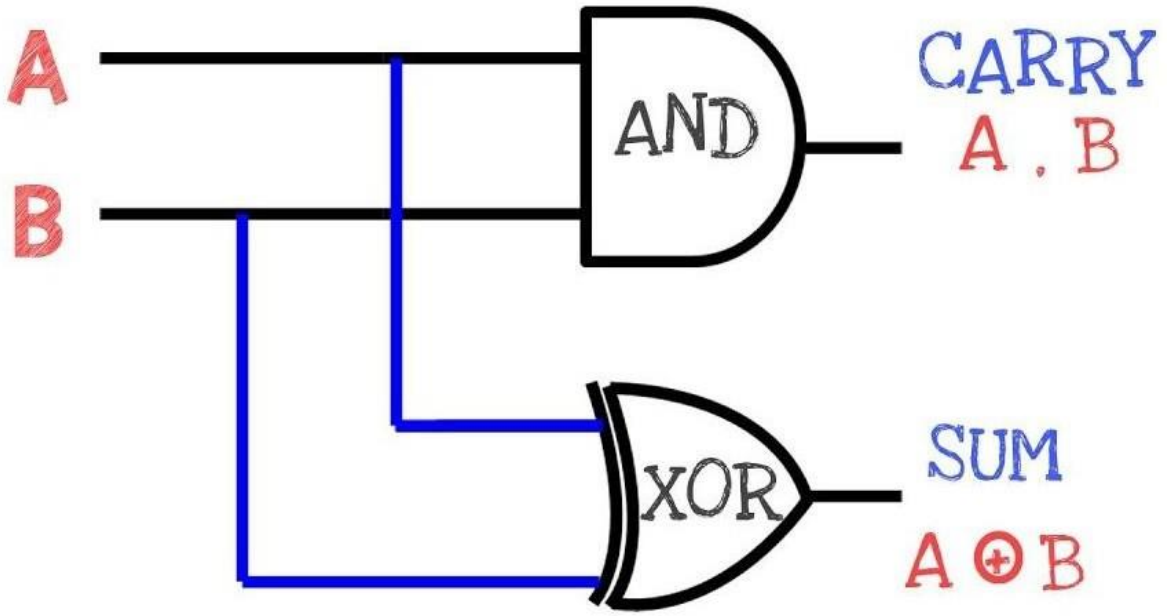
$S = A \oplus B$

Checkboard Configuration

Half Adder Designing

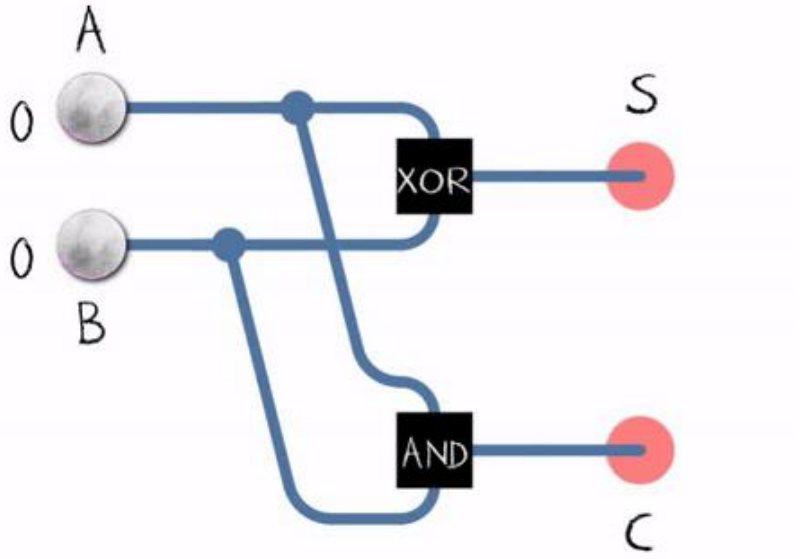
Step 4

- Draw the logic diagram.
- The implementation of half adder using 1 XOR gate and 1 AND gate is as shown below.



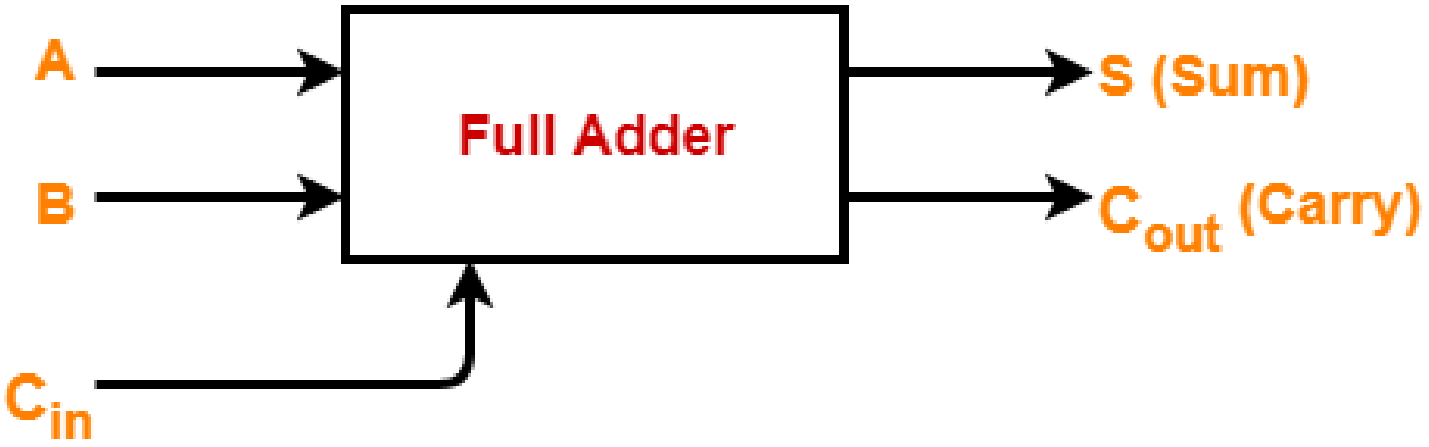
Limitation of Half Adder

- Half adders have no scope of adding the carry bit resulting from the addition of previous bits.
- This is a major drawback of half adders.
- This is because real time scenarios involve adding the multiple number of bits which can not be accomplished using half adders.
- To overcome this drawback, full adder comes into play.



Full Adder

- Full Adder is a combinational logic circuit.
- It is used for the purpose of adding two single bit numbers with a carry.
- Thus, full adder has the ability to perform the addition of three bits.
- Full adder contains 3 inputs and 2 outputs (sum and carry) as shown-



Full Adder Designing

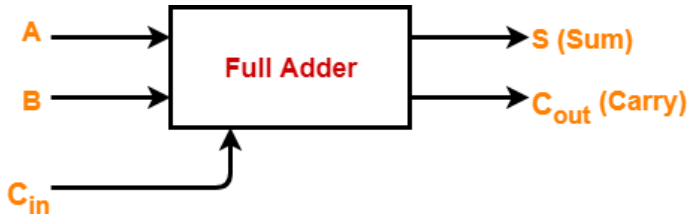
Step 1

- Identify the input and output variables-
  - Input variables = A, B, C_{in} (either 0 or 1)
  - Output variables = S, C_{out} where

S = Sum and
C_{out} = Carry

Step 2

Draw the truth table



Inputs			Outputs	
A	B	C_{in}	C_{out} (Carry)	S (Sum)
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Truth Table

Full Adder Designing

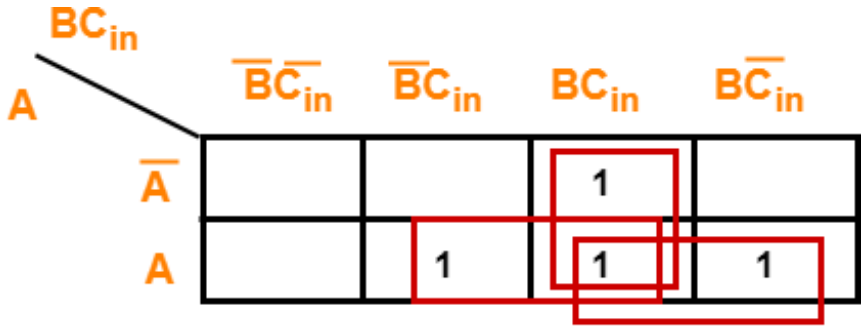
Step 3

- Draw K-maps using the above truth table and determine the simplified Boolean expressions



$$S = A \oplus B \oplus C_{in}$$

Checkboard Configuration



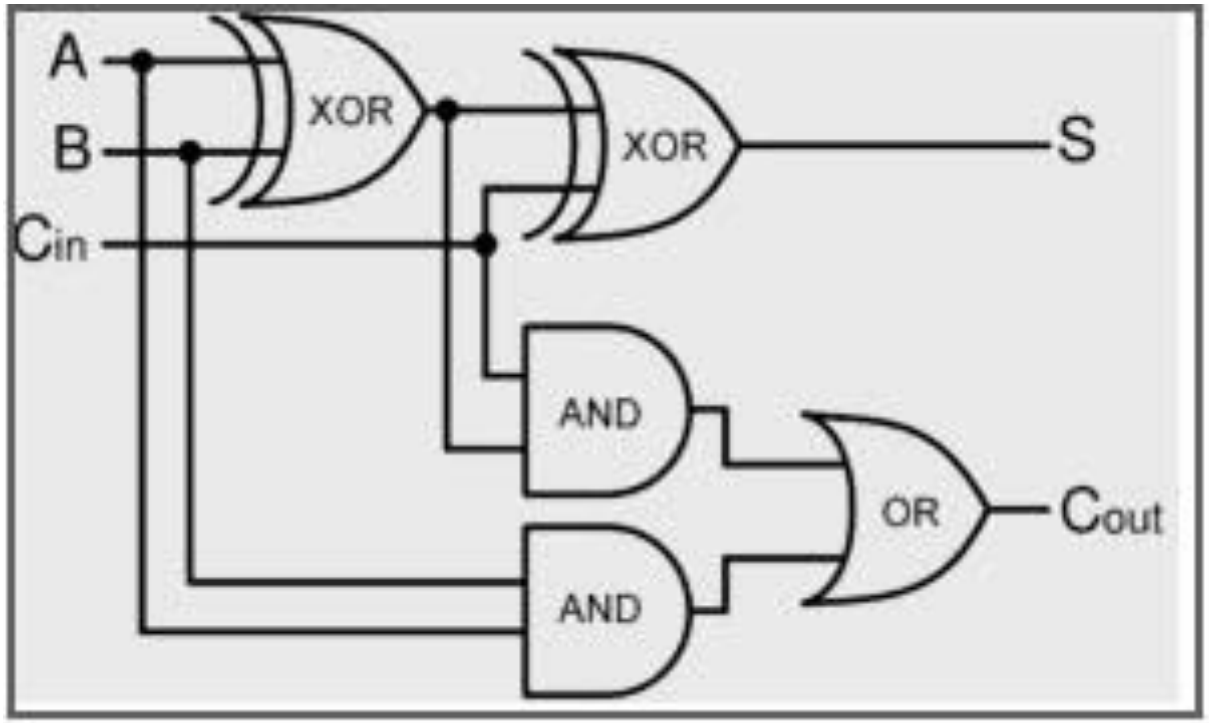
$$C_{out} = AB + BC_{in} + C_{in}A$$

Cout = AB + Cin (A ⊕ B)

Full Adder Designing

Step 4

- Draw the logic diagram.
- The implementation of full adder using 2 XOR gate, 2 AND gates and 1 OR gate is as shown below-

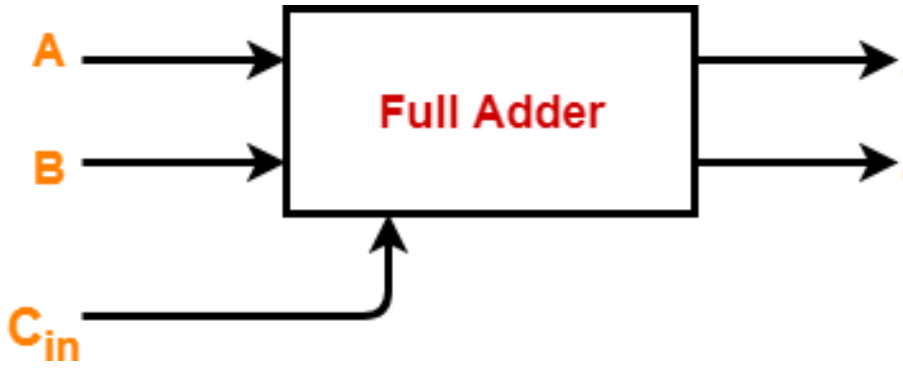


Implementing Full Adder using 2 Half Adders



$S = (A \oplus B)$

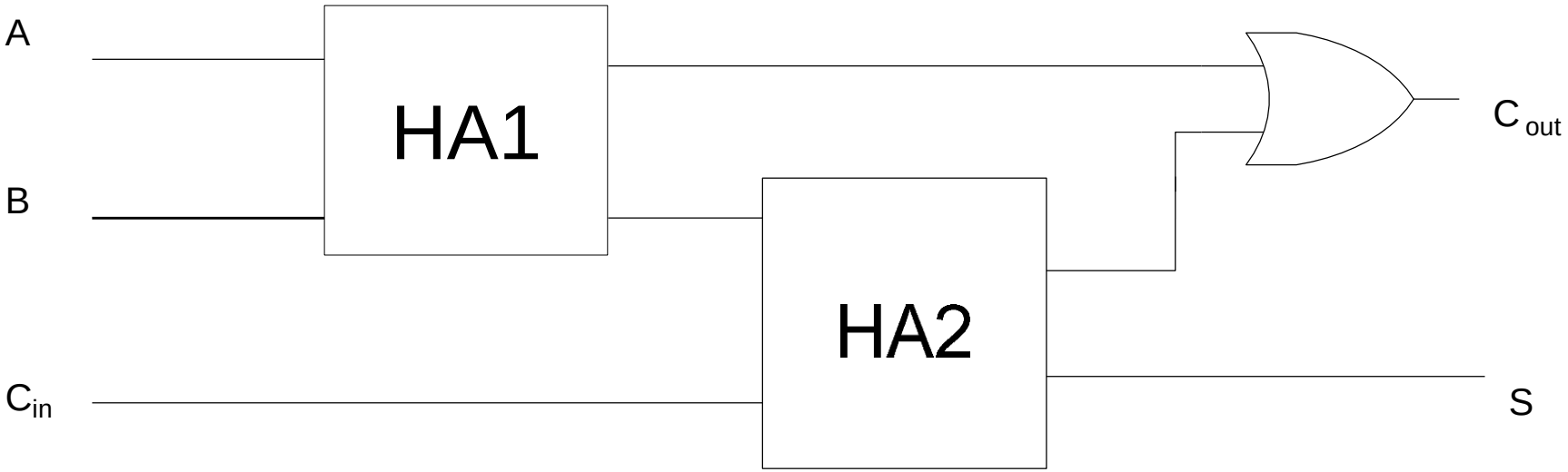
$C = AB$



$S = (A \oplus B \oplus Cin)$

$Cout = AB + Cin\ (A \oplus B)$

Implementing Full Adder using 2 Half Adders

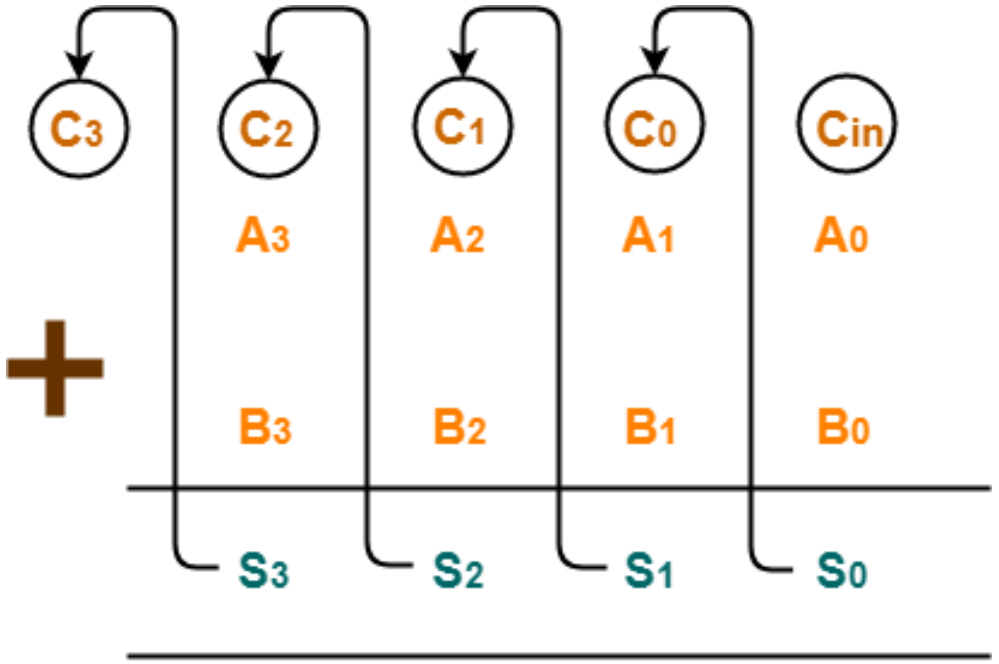


Ripple Carry Adder

- It is used for the purpose of adding two n-bit binary numbers.
- It requires n full adders in its circuit for adding two n-bit binary numbers.
- It is also known as n-bit parallel adder.
- Carry-out of each full adder is the carry in of the succeeding next most significant full adder.
- Each carry bit gets rippled into the next stage

4 bit Ripple Carry Adder

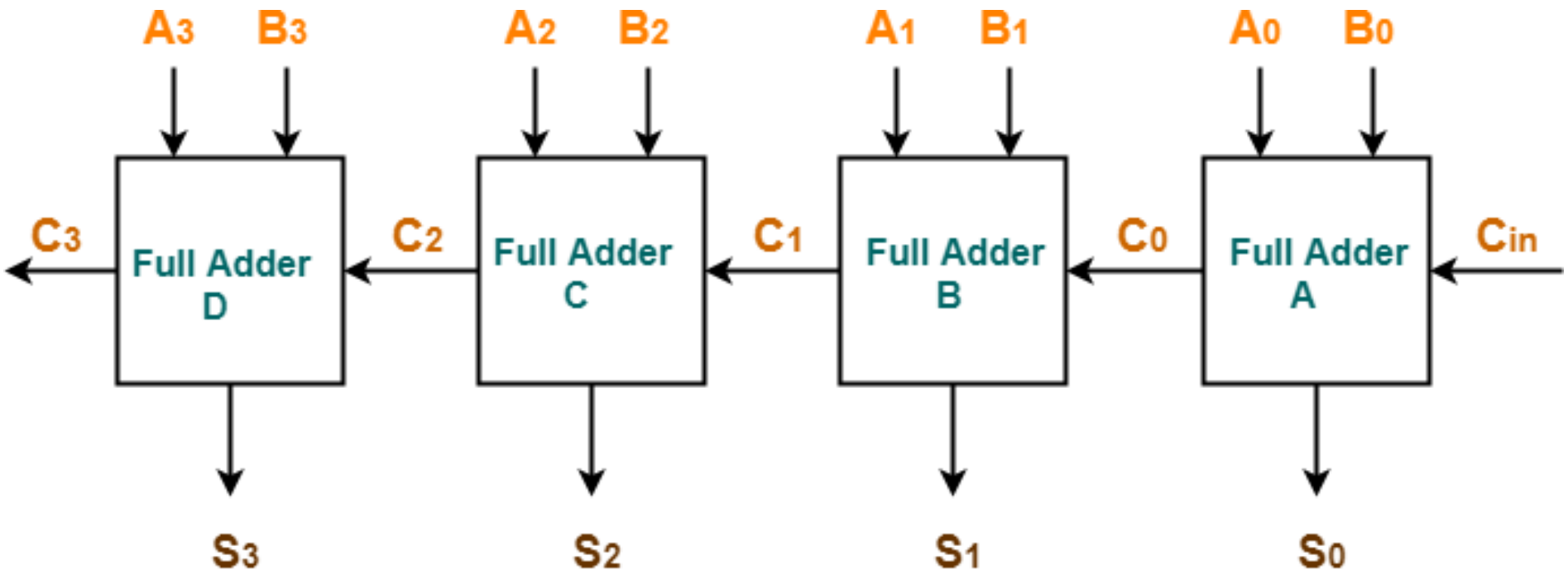
- 4-bit ripple carry adder is used for the purpose of adding two 4-bit binary numbers.



Adding two 4-bit Numbers

4-bit Ripple Carry Adder

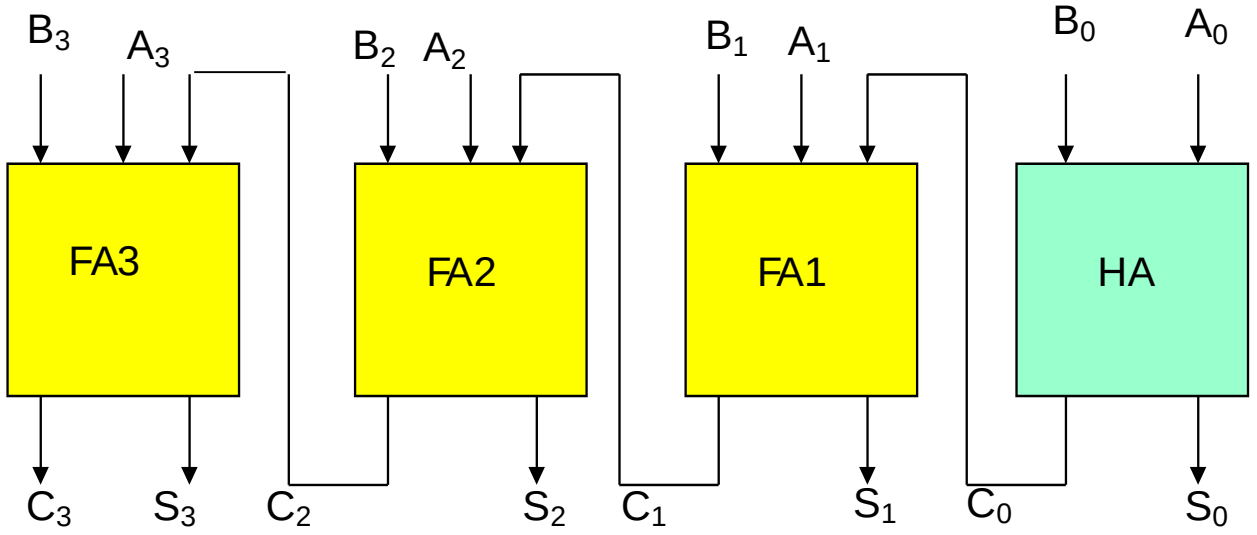
Using ripple carry adder, this addition is carried out as shown by the following logic diagram



4-bit Ripple Carry Adder

4-bit Ripple Carry Adder using Half Adder

- To add A0 and B0 you can use a HA because there is no input carry coming from the previous stage.
- For other stages FAs can be used.



Thank you

